Energy MBA • Energy Markets Quantitative Analysis (EMQA)
Prof. dr. Daniel Traian PELE
Bucharest University of Economic Studies • Romanian Academy, Institute for Economic Forecasting

Assignment 2: Risk, contracts and forecast-driven decisions for your role

Price–volume risk, hedging and PPAs, flexibility and storage, machine-learning forecasts — lectures 4–7

Instructor to fill in: Deadline: **TODO** • Weight in the final grade: **TODO** • Individual or teams of 2: **TODO** • Oral defence slots: **TODO** • Late policy: **TODO**. Submit on Moodle.

The situation

You are the analyst who advises the head of trading of a Romanian energy company. Pick the role closest to your own job (Table 1): a battery-storage operator, a hydro producer facing a dry year, a solar producer choosing between merchant sale, a power purchase agreement (PPA) and a contract for difference (CfD), or a supplier/industrial consumer buying on the day-ahead market (PZU). Every day you decide, before gate closure (12:00 CET on day $d-1$), what to bid or nominate for day d; every year you decide how much to lock in by contract. Your head of trading asks: (1) How much money do better forecasts actually make or save, compared with a simple rule and with perfect foresight? (2) Where does our cash-flow risk come from — price, volume, profile, imbalances or the regional basis — and how much can we lose in a bad month? (3) Which contract or hedge should we sign, and how much? You answer with a reproducible notebook, a decision ledger and a one-page recommendation.

1 What you submit

D1 Notebook `A2_<ID>_<ROLE>.ipynb` from the starter `A2_starter_EN.ipynb`; runs with *Restart & Run All* in at most 45 minutes.

D2 Numbers file `my_numbers_A2_<ID>_<ROLE>_personal.json` and the **decision ledger** as CSV (`ledger_<ID>.csv`), both written by the notebook.

D3 Memo (PDF, 11 pt, at most 2 pages): page 1 = recommendation; page 2 = appendix (tables, ledger excerpt, hand reconciliation). Structure in Task 8.

D4 Oral defence (about 10 minutes) on one day of your ledger and one adverse scenario chosen by the instructor.

2 Role and personal parameters

Choose **one** role. Use public data only: the volumes are *normalised public profiles* built from national ENTSO-E data (European Network of Transmission System Operators for Electricity) (via energy-charts.info), not the portfolio of any real company. Never use or disclose confidential company data. Then run `python make_assignment.py --id <ID> --assignment 2 --role <B|H|R|S>` (or the first cell of the starter). Your test window (365 days starting between January and November 2025), your role settings and your three `check_days` depend on your ID.

Three periods, in this order (all dates in your parameter block):

- ⊡ (1) *estimation*: everything before `selection_start`; the forecast models are recalibrated on a rolling `lear_window`, as in Assignment 1, and `burn_in_days` = 112 forecast days before `selection_start` feed the HS/CP errors;

- ⊡ (2) *selection*: `selection_start`–`selection_end`, the 120 days just before the test; choose here, and only here, your rule, α, w^*, K, the PPA prices and the toll;
- ⊡ (3) *test*: `test_start`–`test_end` (365 days); all choices are frozen; report P&L, VaR breaches and every other result on this period only;
- ⊡ `ea.selection_check(select_fn, P, L, R, p)` must pass: it hides the test window and checks that your choices do not change.

Example (DEMO-0001): selection 3 June – 30 September 2025, test 1 October 2025 – 30 September 2026, forecasts from 11 February 2025. Common parameters: `test_start`, `test_end` (365 days); `selection_start`, `selection_end` (120 days); `burn_in_days` = 112; `lear_window` (364/546/728) and `error_window` W (56/84/112) as in Assignment 1; `var_window` = 250 and `alphas` = [0.95, 0.99] (Task 5); `risk_knob` = α values of the quantile rule (Task 3); three `check_days` for the hand reconciliation. Battery: `power_mw` = 1, `cycles_per_day` = 1. Units: tables in kEUR (battery: EUR); the ledger always in EUR. Keep your notebook in `repo/assignments/A2/`.

Table 1: Roles, volumes (hour h of day d, MWh per hour = mean MW) and personal settings

Role (who should pick it)	Volume and baseline exposure X_d	Personal settings
B Battery operator (storage and renewable developers, suppliers with storage)	1 MW battery, energy E MWh, round-trip efficiency η, degradation cost c EUR per MWh discharged; at most one cycle per day, empty at midnight. X_d = daily profit of the *naive-forecast* strategy (Task 3), EUR.	`energy_mwh` $\in \{1, 2\}$, `efficiency_rt`, `degradation_eur_mwh`
H Hydro producer, dry year (hydro producers; gas-plant owners may also pick it)	$Q_{d,h} = \phi\,(\text{reservoir}_{d,h} + \text{run-of-river}_{d,h})$, national ENTSO-E hydro output times your fleet share ϕ. $X_d = \sum_h Q_{d,h} P_{d,h}/1000$ (thousand EUR, kEUR).	`fleet_share` ϕ, `flex_share`, `pmax_factor`
R Solar producer (renewable developers, PPA/CfD desks)	$Q_{d,h} = C \cdot S_{d,h}/\bar{S}_d$, national solar output S divided by its trailing 365-day peak $\bar{S}_d$ up to day $d-2$, times your park size C MWp. $X_d = \sum_h Q_{d,h} P_{d,h}/1000$ kEUR.	`pv_mwp`, `ppa_hedge`, `curtail_threshold_eur`
S Supplier or industrial consumer (suppliers, industry; nuclear and grid staff buying replacement or loss energy)	$Q_{d,h} = \bar{Q} \cdot L_{d,h}/\bar{L}_d$, national load divided by its trailing 365-day mean up to day $d-2$, times your average load $\bar{Q}$ MW. $X_d = -\sum_h Q_{d,h} P_{d,h}/1000$ kEUR (a cost is a negative cash flow).	`avg_load_mw`, `flex_share`, `max_load_factor`

Same data rules as Assignment 1: CET hours `h00`–`h23`; hourly values (mean of the quarter-hours after 1 October 2025); and the gate-closure information set (12:00 CET on $d-1$), enforced by `ea.leakage_check`:

- ⊡ prices: every day up to $d-1$ (the $d-1$ auction cleared on $d-2$);
- ⊡ realised quantities (load, wind, solar, hydro, your volumes and cash flows that depend on them): only up to day $d-2$, the last metered day; proxies for day d come from $d-2$, $d-3$ and $d-7$;
- ⊡ fuels and carbon: the last close before 12:00 on $d-1$, in practice the close of $d-2$;
- ⊡ never data of day d, nor realised quantities of $d-1$, in a decision for day d.

Read Section 3 of the Assignment 1 brief again if needed.

3 Tasks

Task 1 — Exposure profile and market context [8 points]

(a) Monthly table from January 2023 to the end of your test window: volume (MWh), cash flow (EUR), base price $\bar{P}_m$ (mean of all hours), capture price $\mathrm{CP}_m = \sum Q_{d,h} P_{d,h} / \sum Q_{d,h}$ and capture rate $\mathrm{CP}_m / \bar{P}_m$ (B: oracle and naive profit per month instead). (b) Role facts on the test window: H — volume of the wet year 2023 vs the dry year 2025; R — volume produced in negative-price hours; S — the profile cost premium $\mathrm{CP} - \bar{P}$; B — mean daily max–min price spread. (c) **Clean spark spread** (CSS) of a gas plant for each month of your window, as context for the marginal price setter:

$$\mathrm{CSS}_m = \bar{P}_m^{\mathrm{peak}} - \frac{\mathrm{TTF}_m}{\eta_g} - \frac{e_g}{\eta_g}\,\mathrm{EUA}_m, \qquad \eta_g = 0.55, \quad e_g = 0.202\ \mathrm{tCO_2/MWh_{th}}, \tag{1}$$

where $\bar{P}_m^{\mathrm{peak}}$ is the mean price of hours h08–h19 on weekdays (base CSS: use the base price $\bar{P}_m$ instead), TTF_m the monthly gas price at the Dutch Title Transfer Facility hub (EUR/MWh, World Bank, `ea.ttf_monthly()`), EUA_m the monthly mean price of an EU emission allowance (EUA; auctions of the European Energy Exchange, EEX, until 2025; for 2026 the price of a physically backed exchange-traded commodity, ETC, as proxy, about 4% below auction prices: mention the level break; `ea.eua_daily()`, columns `eua_eex_auction_eur_t` then `eua_etc_co2l_eur_per_unit`), η_g the plant efficiency and e_g the emission factor of gas.

Show: monthly table; role-facts table; CSS table (base and peak); two sentences on what the numbers mean for your role.

Task 2 — Probabilistic forecasts: LEAR vs a machine-learning model [8 points]

Forecast every hour of burn-in + selection + test with: Naive and LEAR (exactly as in Assignment 1, your `lear_window`), and **one** machine-learning (ML) model of your choice: a multi-output neural network (`MLPRegressor`, 24 outputs, the 247 LEAR inputs, at least 2 random seeds averaged) [3], or gradient-boosted trees / a random forest (one model per hour). Rolling calibration on your `lear_window`, recalibrated at least every 7 days, with the same 247 inputs; fix `random_state` so that `ea.leakage_check` is deterministic. Add the equal-weight average of LEAR and ML. If you use early stopping, validate on the *last* days of the calibration window: `MLPRegressor(early_stopping=True)` and `HistGradientBoostingRegressor` (by default above 10,000 rows) draw a *random* validation subset, which mixes later days into the training data. Turn each into 99 quantiles with HS and CP (Assignment 1, Task 6; your `error_window`). Report MAE, RMSE, APS, coverage of the 50% and 90% intervals, and DM tests (HAC) for Naive vs LEAR, LEAR vs ML, LEAR vs average.

Optional bonus (+5, total capped at 100): a zero-shot foundation model (e.g. Chronos-2 or TabPFN) [6]. Record the exact model version and its pre-training cutoff date; discuss whether the pre-training data may already contain your test window (data contamination).

Show: table T2 (rows = model + HS/CP; columns MAE, RMSE, APS, cov50, cov90); DM p-values; one sentence: does ML beat LEAR on your window, and is it significant?

Task 3 — The daily decision, the decision ledger and its economic value [20 points]

Apply your role's rule every selection and every test day with these signals: **oracle** (actual prices — an upper bound, never achievable), the **naive**, **LEAR** and **ML** point forecasts, and (B and R only) the **quantile rule** with the **LEAR + CP** quantiles for each α in `risk_knob` $= \{0.5, \ldots, 0.9\}$ (prescribed). **Choose your recommended signal (and α) by the highest total cash flow on the selection period**, freeze it, and report its test value next to the other signals; the instructor's `MODE="latest"` re-run is a second unseen check. Realised cash flows always use actual prices $P_{d,h}$. $\hat{q}^{\alpha}_{d,h}$ is the α-quantile forecast.

B Choose a charging block of E consecutive hours starting at hour t_c and a discharging block starting at $t_x \geq t_c + E$, with $t_x + E \leq 24$, that maximise the signal value $V = \eta \sum_{k=0}^{E-1} s^{\text{sell}}_{d,t_x+k} - \sum_{k=0}^{E-1} s^{\text{buy}}_{d,t_c+k} - c\,\eta E$; trade only if $V > 0$ (price-taker orders, both legs executed). Point rule: $s^{\text{buy}} = s^{\text{sell}} = \hat{P}$. Quantile rule: $s^{\text{buy}} = \hat{q}^{\alpha}$ (pessimistic buy price), $s^{\text{sell}} = \hat{q}^{1-\alpha}$ (pessimistic sell price). Profit $\pi_d = \eta \sum_k P_{d,t_x+k} - \sum_k P_{d,t_c+k} - c\,\eta E$ (0 on days without trade). Baseline for B: no operation (cash flow 0). This simplifies [5, 4].

H The day's reservoir energy is planned at gate closure from the last metered day, $E_d = \phi \sum_h \text{reservoir}_{d-2,h}$ (persistence; water management fixes the release before gate closure). A share $1 - f$ (`flex_share` $= f$) runs flat (minimum flow); the flexible energy fE_d goes to the hours with the highest signal, at most (`pmax_factor` $\cdot\, E_d/24$) MW per hour in total. Run-of-river is not dispatchable. Baseline: the whole E_d flat. Value = revenue of the rule − revenue of the baseline.

R Curtail hour h if the signal is below your threshold θ (`curtail_threshold_eur`). Point rule: $\hat{P}_{d,h} < \theta$. Quantile rule: $\hat{q}^{\alpha}_{d,h} < \theta$, i.e. curtail only if the price is below θ with probability at least α. Baseline: never curtail.

S The shift is planned on the profile known at gate closure, $\tilde{Q}_{d,h} = Q_{d-2,h}$: every hour gives up $f\tilde{Q}_{d,h}$, and the flexible energy $f \sum_h \tilde{Q}_{d,h}$ is placed in the hours with the lowest signal, at most `max_load_factor`$\cdot\bar{Q}$ MW per hour in total (on the plan); the realised volume is $Q_{d,h} - f\tilde{Q}_{d,h} + \text{added}_h$. Baseline: no shifting. Value = baseline cost − cost of the rule.

Capture of the oracle = value of the rule / value of the oracle (B: profit / oracle profit). **ES95 and daily CFaR95 in T3, T4 and T6**: $\text{CFaR}^{95} = \overline{X} - Q_{0.05}(X)$ and $\text{ES}^{95} = \overline{X}$ − mean of the worst 5% of X, positive numbers, over the days of the test window. **Decision ledger** (your chosen rule only; amounts in EUR): one row per test day with the columns `date, info_cutoff, forecast_used, decision, volume_MWh, settlement_price_summary, cash_flow_EUR, baseline_cash_flow_EUR, difference_EUR`.

Show: table T3 (rows = signals; columns total cash flow, value vs baseline, days traded/changed, worst day, ES95 of the daily cash flow (Expected Shortfall, defined in Task 5), capture of oracle in %); the ledger CSV; your choice of rule in one sentence.

Task 4 — Energy risk: price, volume, profile, imbalance and basis [12 points]

Energy cash flows are price × volume, and the two move together. Measure the risks that matter for your role, on real data:

(a) **Price–volume decomposition** (H, R, S). For each month m of your window compare with the same month one year earlier ($m - 12$), with Q_m the monthly volume and CP_m the capture price:

$$\underbrace{Q_m \text{CP}_m - Q_{m-12}\text{CP}_{m-12}}_{\Delta\text{CF}_m} = \underbrace{\Delta Q_m\, \text{CP}_{m-12}}_{\text{volume effect}} + \underbrace{Q_{m-12}\, \Delta\text{CP}_m}_{\text{price effect}} + \underbrace{\Delta Q_m\, \Delta\text{CP}_m}_{\text{joint effect}}, \tag{2}$$

where $\Delta Q_m = Q_m - Q_{m-12}$ and $\Delta\text{CP}_m = \text{CP}_m - \text{CP}_{m-12}$. Also report the correlation between daily volume and the daily base price on the test window. A negative correlation for a producer is a *natural hedge* (dry year: less water, higher prices); a positive one for a buyer means you consume more when power is expensive.

(b) **Role risk.** H: volume and revenue of the wet year 2023 vs the dry year 2025 (the dry-year shortfall). R: capture rate per year 2023–2026 (solar cannibalisation) and the share of your volume produced at negative prices. S: the profile cost premium $\text{CP} - \bar{P}$ in EUR/MWh (what a flat base-load contract does not cover). B: mean and 5% quantile of the daily max–min spread, and the number of days your chosen rule lost money.

(c) **Imbalance exposure** (H: run-of-river; R; S). Any difference between the volume you nominate day-ahead and the volume you deliver is settled on the balancing market (*piața de echilibrare*) at the imbalance price. Our public data contain no imbalance prices, so measure the *volume*: with the

persistence forecast $\tilde{Q}_{d,h} = Q_{d-2,h}$ (the last metered day at gate closure), the imbalance volume is $\sum_{d,h} |Q_{d,h} - \tilde{Q}_{d,h}|$ MWh per year; report the cost per 10 EUR/MWh of average imbalance-price spread. Do not invent imbalance prices.

(d) **Regional basis RO–HU.** On the test window: mean of $P^{\mathrm{RO}}_{d,h} - P^{\mathrm{HU}}_{d,h}$, its 5% and 95% quantiles, and the share of hours with $|P^{\mathrm{RO}} - P^{\mathrm{HU}}| > 1$ EUR/MWh (market decoupling, e.g. summer 2024). This is the basis risk of hedging Romanian exposure with a contract settled on a neighbouring price (Task 6).

(e) **One summary of the cash-flow distribution**: monthly Cash-Flow-at-Risk from January 2023, $\mathrm{CFaR}^{95} = \overline{X}_m - Q_{0.05}(X_m)$ (X_m = monthly sum of X_d), and on the test window the daily CFaR^{95} and the mean of the worst 5% of days below the mean (an Expected-Shortfall-type number). The formal VaR/ES module with backtests is Task 5.

Show: decomposition table (monthly, kEUR); a role-risk table; imbalance volume and cost per 10 EUR/MWh; RO–HU basis table and a figure of the daily mean spread; CFaR numbers; two sentences: which risk dominates for you?

Task 5 — VaR and ES of your cash flow, with backtests [16 points]

One module, applied to *your role's* daily cash flow X_d (Table 1; a procurement cost for S), not to share prices. Use X_d from 2022 to the end of your window. For each test day d and $\alpha \in \{0.95, 0.99\}$, forecast the lower tail of X_d with the information available at gate closure only, i.e. cash flows up to day $d-2$ (a two-day-ahead forecast):

$$q^{\alpha}_d = Q_{1-\alpha}(X_d \mid \mathcal{F}_{d-2}), \qquad \mathrm{VaR}^{\alpha}_d = \mu_d - q^{\alpha}_d \ (\geq 0), \qquad \mathrm{ES}^{\alpha}_d = \mu_d - \mathbb{E}[X_d \mid X_d < q^{\alpha}_d, \mathcal{F}_{d-2}], \quad (3)$$

where $\mathcal{F}_{d-2}$ is the information up to day $d-2$, $Q_{1-\alpha}$ the $(1-\alpha)$-quantile and μ_d the forecast mean (HS and Normal: mean of the 250 days up to $d-2$; MC: the two-step-ahead conditional mean of the AR–GARCH model, `arch_model(X, mean="AR", lags=[1,7], vol="GARCH", dist="t")`). VaR (Value-at-Risk) and ES (Expected Shortfall) are *positive* amounts of money below the expected cash flow, as in Cash-Flow-at-Risk (lecture 4). A **breach** is $I_d = \mathbf{1}\{X_d < q^{\alpha}_d\}$. Three methods: (i) historical simulation (HS) on the last 250 days; (ii) Normal with the mean and standard deviation of the last 250 days; (iii) Monte Carlo (MC) from an AR(1,7)–GARCH(1,1) model with Student-t errors (AR lags 1 and 7 in the mean), re-estimated every 21 days on the last 750 days, 20 000 draws. Backtest with the Kupiec test (Assignment 1, $\mathrm{LR}_{\mathrm{UC}}$, now with $p_0 = 1-\alpha$) and the Christoffersen independence test [1]:

$$\mathrm{LR}_{\mathrm{ind}} = -2 \ln \frac{(1-\hat{\pi})^{n_{00}+n_{10}} \hat{\pi}^{\,n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1-\hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \sim \chi^2_1, \quad (4)$$

where n_{ij} counts days with $I_{d-1} = i$ and $I_d = j$, $\hat{\pi}_{01} = n_{01}/(n_{00}+n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10}+n_{11})$, $\hat{\pi} = (n_{01}+n_{11})/(n_{00}+n_{01}+n_{10}+n_{11})$. **ES check with its uncertainty**: on the breach days compare the realised shortfall $\mu_d - X_d$ with the forecast ES^{α}_d; report the mean gap and a 90% bootstrap interval (resample the breach days 2 000 times). With 365 days and $\alpha = 0.99$ you expect only 3–4 breaches: the tests have little power and ES comparisons are very imprecise [2]. Say so; do not rank methods on two or three tail days.

Show: table T5 (rows = 3 methods × 2 levels; columns breaches, expected, Kupiec p, Christoffersen p, mean VaR, mean ES, ES gap with 90% interval); figure of X_d with the three 99% quantiles; the method you would use and why.

Task 6 — Contract or hedge design [12 points]

No invented forward prices. Our free data contain no executable forward or futures history. EEX lists Romanian base and peak futures and OPCOM runs bilateral contract markets, but liquidity and price history are not public in our data set. Therefore you compare structures at **equal mean**: the

contract price K is set to the value that gives the same average cash flow as the unhedged position over the *selection* period, and is then frozen for the test. K is a comparison device, not a forecast and not a market quote. Then only the risk differs. Plan volume $\hat{Q}_d$ = mean daily volume of the same calendar month one year earlier (known when the contract is signed).

H, S Base-load swap on a share $w \in \{0, 0.25, 0.5, 0.75, 1\}$ of the plan, all amounts in EUR: $X_d^{(w)} = X_d + s\,w\,\hat{Q}_d(K - \bar{P}_d)$ with $s = +1$ (H sells) or $s = -1$ (S buys), $\bar{P}_d$ the daily base price and $\hat{Q}_d$ in MWh per day. Minimum-variance ratio $w^* = s\,\mathrm{Cov}(X_d, \hat{Q}_d\bar{P}_d)/\mathrm{Var}(\hat{Q}_d\bar{P}_d)$ (lecture 5), estimated on the selection period and capped at 1. Report on the test period the mean, standard deviation, daily CFaR^{95}, ES^{95} and worst day for each w and for w^*, the correlation between volume and price (the "natural hedge" of a hydro producer), and the basis that a base-load contract leaves open (profile and volume). Repeat the table with the swap settled on the *Hungarian* daily base price (a cross-hedge): the extra risk is the RO–HU basis of Task 4(d).

R Compare on the test period, for a share s (`ppa_hedge`): merchant; pay-as-produced PPA $(1 - s)\sum_h QP + sK_1\sum_h Q$ with K_1 = capture price of the selection period; base-load PPA $\sum_h QP + s\bar{Q}\sum_h(K_2 - P_{d,h})$ with $\bar{Q}$ the mean hourly output over all hours of the selection period (night included, no curtailment) and K_2 the mean base price of the selection period; two-sided CfD with hourly reference price and no payment in negative hours, $\sum_h QP + s\sum_{h:P\geq 0} Q(K_3 - P)$, with K_3 the capture price over non-negative hours of the selection period (all prices and volumes of the contracts are fixed on the selection period). Discuss capture-price risk, profile/volume basis and negative-price clauses (Romania's 2024–2025 CfD scheme; check its official reference-price definition).

B The risk–return frontier of the quantile rule: total profit, ES^{95} of the daily profit, worst day and days traded for each α, on the selection period (choose α there) and on the test period (report). Then price a fixed **tolling** contract (an offtaker pays a fixed EUR/MW/year and takes the merchant risk): the toll is the annualised profit of your chosen rule on the selection period; report the offtaker's result on the test period.

Show: table T6 for your role; your recommended share or structure in one sentence, with its risk reduction in EUR.

Task 7 — Robustness and an adverse scenario [8 points]

(a) Value of your chosen rule per quarter: is it stable or earned in one quarter? (b) Role-specific adverse scenario (a stress test, not a forecast): B — efficiency −5 percentage points and degradation cost doubled, applied to the LEAR point rule; H — 30% less water (all hydro volume ×0.7, prices unchanged) on the baseline exposure, compared with the real 2023 vs 2025 volumes; R — every negative hour clears at the new floor −600 EUR/MWh (from 29 May 2026), merchant revenue with and without curtailment of those hours; S — the 10 days with the highest daily base price get all 24 hourly prices ×1.5, baseline cost. Report the effect in EUR and on the daily CFaR^{95}.

Task 8 — The memo: recommendation, reconciliation, your company [16 points]

Page 1 (at most 450 words): (a) the decision in the first two lines — which daily rule and which contract/hedge share, from when; (b) the money: value vs baseline and capture of oracle, the dominant energy risk and CFaR before and after the hedge; (c) what failed (backtest, quarter, scenario) and what you would change; (d) **How this applies in my company** (at most 150 words, public information only: e.g. PZU bidding, imbalance-cost exposure on the balancing market, bilateral or PPA pricing, battery dispatch). **Page 2 (appendix)**: parameter block; excerpts of T2–T6; ten ledger rows; and the **hand reconciliation** of your three `check_days`: recompute the cash flow of each day by hand from the hourly prices and volumes (B: buy and sell prices of the chosen hours; H, S: the three largest hourly changes vs baseline; R: the curtailed hours, or "no curtailment").

4 Worked examples (real data)

(1) One battery day in the ledger (1 MW / 1 MWh, $\eta = 0.90$, $c = 5$ EUR/MWh). Day 23 April 2026, information cut-off 22 April 12:00 CET. The LEAR forecast puts the cheapest hour at h13 and the most expensive later hour at h20; signal value $V = 118.2 > 0$, so the battery trades. Actual prices: $P_{h13} = -0.39$, $P_{h20} = 186.31$ EUR/MWh. Profit $= 0.9 \times 186.31 - (-0.39) - 5 \times 0.9 \times 1 = 167.68 + 0.39 - 4.50 = 163.57$ EUR. Buying at a negative price *earns* 0.39 EUR.

(2) Price–volume decomposition, one month (illustrative round numbers). Last year: $Q = 100\,000$ MWh at CP = 100 EUR/MWh; this year $Q = 80\,000$ MWh at CP = 130. $\Delta\text{CF} = 10.4 - 10.0 = +0.4$ million EUR = volume effect $(-20\,000 \times 100 = -2.0)$ + price effect $(100\,000 \times 30 = +3.0)$ + joint effect $(-20\,000 \times 30 = -0.6)$ million EUR. A dry month cost 2 million EUR of volume, but higher prices more than compensated: the natural hedge at work.

(3) One hedged day (H, seller; illustrative numbers). Plan $\hat{Q}_d = 10\,000$ MWh, $K = 124.49$ EUR/MWh, base price $\bar{P}_d = 150$, $w = 0.5$: hedge payoff $= 0.5 \times 10\,000 \times (124.49 - 150) = -127\,550$ EUR. The swap loses when prices are high, exactly when the unhedged revenue is high: that is the point of a hedge.

5 Rules, pitfalls, checklist

Rules R1–R6 of Assignment 1 apply unchanged (end-to-end run, no hard-coded dates and `MODE="latest"`, data version, no look-ahead, every memo number in `MY_NUMBERS`, artificial-intelligence (AI) tools for code only, with disclosure). The perfect-foresight (oracle) result must be labelled "upper bound"; presenting it as achievable costs 5 points.

- ⊡ **Sign conventions**: X_d is a cash flow (a cost is negative); CFaR is a positive amount below the mean. For a buyer, the bad tail is a *higher* cost. Write the convention once and keep it.
- ⊡ **Price risk only**: ignoring volume, profile and imbalance risk, or treating price and volume as independent.
- ⊡ **Look-ahead**: using day d prices to choose battery hours, using the realised monthly volume as "plan", using the realised volume of day d as its own forecast; choosing the rule, α, w^* or K on the test window.
- ⊡ **Infeasible schedules**: discharging before charging, more than one cycle per day, ignoring efficiency or degradation, exceeding the power limit.
- ⊡ **Double counting**: adding balancing-reserve revenue (aFRR/mFRR, automatic/manual frequency restoration reserves) on top of arbitrage with the same MW. Reserves are an optional discussion only.
- ⊡ **Log-returns of cash flows or prices** that can be zero or negative; 15-minute vs hourly mixing; UTC vs CET.
- ⊡ **Invented hedge prices** or forward curves; "the hedge earned X" without saying that K is the equal-mean device.

☐ Role chosen and justified in one line; parameter block in notebook and memo.

☐ *Restart & Run All* and `MODE="latest"` both run; data version printed; leakage check passed for LEAR and ML.

☐ T1–T6, ledger CSV, energy-risk tables, VaR/ES backtests, adverse scenario, figures with units.

☐ Memo ≤ 2 pages, decision first, oracle labelled, hand reconciliation of 3 days, "AI use" cell, no confidential data.

6 Frequently asked questions

My company is not in the list (nuclear, gas, grid operator, trader).
Pick the closest mechanism: a nuclear producer buying replacement energy during an outage, or a grid operator buying losses, is an S buyer; a gas-plant owner with limited fuel or a flexible plant is close to H; a trader with storage or spread positions is close to B. Explain the mapping in your company section.

Can I use the intraday market or imbalance prices?
Not in the graded part: Romanian intraday (ID1-type indices) and imbalance prices are not in our public data

set. You may discuss them qualitatively in the company section.

The battery works on 15-minute prices in reality.
True since October 2025. The graded version is hourly; a 15-minute variant is a welcome extra figure (no extra points).

Why is the hydro energy budget taken from actual data?
Daily water release is decided by water management and is outside the price decision. Treat E_d as given; your decision is *when* to use it. Say this limitation in the memo.

Why no imbalance prices?
They are not in our free data set. Measure the imbalance volume and give the cost per 10 EUR/MWh of spread; if your company publishes typical imbalance costs, quote the public figure with its source.

Why do I need data from 2022 for VaR when my test window is 2025–2026?
HS uses the 250 days before each test day and the GARCH model 750 days. All of them are before the forecast day, so there is no leakage.

Only 3 breaches at 99%. Which method is best?
You cannot tell reliably from so few tail days. Report the tests, the ES gap with its interval, and base the choice on the 95% level and on the economics of your role.

w^* is larger than 1. Should I hedge more than 100%?
No: over-hedging creates a speculative position when volume falls. Cap at 1 and explain why $w^* > 1$ (plan volume below actual, volume–price correlation).

The ML model loses to LEAR. Is that a bad result?
No. It is a common finding with short samples [3, 6]. Report it honestly and check whether the average of the two helps.

Which numbers will the instructor check?
The instructor has a reference implementation and recomputes T1, T3, T4, T5 and T6 for your ID and role. Small differences from legitimate choices (e.g. another ML model) are fine if explained.

7 Grading rubric (100 points)

Task	Pts	Excellent (90–100%)	Adequate (50–70%)	Poor (0–30%)
1 Exposure	8	correct profile, capture price, role facts and CSS, interpreted for the role	tables right, interpretation generic	wrong profile or units
2 Forecasts	8	LEAR + one ML, rolling, HS/CP, APS, coverage, HAC DM, honest comparison	point metrics only or no DM	in-sample or leakage
3 Decision, ledger	20	feasible rule, all signals, oracle as upper bound, quantile frontier (B, R), choice frozen on the selection period, `selection_check` passed, complete ledger, value in EUR	one signal missing or ledger incomplete	infeasible schedule, oracle as result, choice made on the test window
4 Energy risk	12	decomposition, role risk, imbalance volume, RO–HU basis, CFaR; dominant risk identified with numbers	two or three items, little interpretation	price risk only, invented imbalance prices
5 VaR/ES	16	3 methods × 2 levels on the role's cash flow, correct signs, Kupiec + Christoffersen, ES gap with interval, small-sample caveat	2 methods or one test, no uncertainty	wrong sign, in-sample VaR, equity returns

Task	Pts	Excellent (90–100%)	Adequate (50–70%)	Poor (0–30%)
6 Contract	12	equal-mean device stated, all structures/ratios, w^* or frontier, RO and HU settlement, basis discussed, clear choice	table without basis discussion	invented forward prices
7 Robustness	8	quarters + adverse scenario quantified and used in the decision	computed, not discussed	absent
8a Decision	8	decision first, money and risk numbers, limits	vague decision	generic text
8b Reconciliation	4	3 days by hand, matches ledger	1–2 days	missing
8c My company	4	concrete, public data only	generic	absent or confidential

Gates and penalties as in Assignment 1 (R1 caps Tasks 1–7 at 50%; 2 points per untraceable number, up to 10; oracle presented as achievable: −5; bonus for a foundation model: up to +5, total capped at 100).

8 Further reading (papers of the course instructor and co-authors)

- ⊡ *Joint Probabilistic Forecasting for Battery Arbitrage in Day-Ahead Power Markets*, SSRN working paper 6872381 (2026), doi:10.2139/ssrn.6872381 — role B.
- ⊡ *Explainable AI-Based Early Warning System for Electricity Price Spikes: Evidence from the Romanian Market*, PICBE 2026, doi:10.2478/picbe-2026-0352 — spikes and risk.
- ⊡ *Can Foundation Models Manage Risk? Zero-Shot VaR and ES Forecasting with Conformal Calibration in CEE Markets*, doi:10.24818/18423264/60.2.26.14 — Task 5 and the bonus.
- ⊡ *Finite-Sample Precision Limits for Expected Shortfall Forecast Comparisons*, *Mathematics* (2026), doi:10.3390/math14132316 — why tail-risk comparisons need many tail observations.
- ⊡ *Evaluating Intervention Effectiveness during Extreme Price Events: Romania's 2024 Energy Crisis*, PICBE 2026, doi:10.2478/picbe-2026-0013.
- ⊡ *Energy Financial Risk Meter: Systemic Risk in the EU Energy Sector*, SSRN working paper 6985301, doi:10.2139/ssrn.6985301.
- ⊡ *Day-ahead Forecasting for Redispatch Measures using Machine Learning Models*, SSRN working paper 7511080, doi:10.2139/ssrn.7511080 — grid operators.
- ⊡ A.-V. Andrei, G. Velev, F.-M. Toma, D.T. Pele, S. Lessmann (2024). *Energy Price Modelling: A Comparative Evaluation of four Generations of Forecasting Methods*, arXiv:2411.03372.

References

[1] P.F. Christoffersen (1998). Evaluating interval forecasts. *International Economic Review* 39(4), 841–862, doi:10.2307/2527341.

[2] *Finite-Sample Precision Limits for Expected Shortfall Forecast Comparisons. Mathematics* 14(13), 2316 (2026), doi:10.3390/math14132316.

[3] J. Lago, G. Marcjasz, B. De Schutter, R. Weron (2021). Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark. *Applied Energy* 293, 116983, doi:10.1016/j.apenergy.2021.116983.

[4] T. Serafin, R. Weron (2025). Loss functions in regression models: Impact on profits and risk in day-ahead electricity trading. *Energy Economics*, 148, 108596, doi:10.1016/j.eneco.2025.108596 (working paper WORMS/24/03, 2024); as presented in R. Weron, IIF Distinguished Lectures on EPF, part II, ISEA 2025, slides 64–66, slides and recordings.

[5] B. Uniejewski, R. Weron (2021). Regularized quantile regression averaging for probabilistic electricity price forecasting. *Energy Economics* 95, 105121, doi:10.1016/j.eneco.2021.105121.

[6] R. Weron (2026). *Machine learning for electricity price forecasting*. Keynote, AIDA 2026, Bucharest, 3–5 June 2026; and IIF Distinguished Lectures on EPF, parts I–II, ISEA 2025, slides and recordings.